FEDERAL SPECIFICATION

HOSE, NON-METALLIC AND HOSE; PREFORMED:
(FOR THE COOLANT SYSTEMS OF AUTOMOTIVE
AND OTHER LIQUID-COOLED ENGINES)

This amendment which forms a part of ZZ-H-428D, dated May 8, 1974, is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

PAGE 3

2.1, under "Military Standards:" add:

"MS51008 - Hose Preformed: Flexible, Engine Coolant.
MS52130 - Hose, Nonmetallic: Straight, Engine Coolant."

2.2, delete in its entirety.

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Table VIII, delete in its entirety and substitute new table VIII.

"Table VIII - Physical properties requirements (types I through V (Cont'd)

Physical properties	Cover requirements							
	Applicable types				Classes			
	I and II	III	IV	V	I	2	3	4
Tensile strength, max., percent	X	---	---		-45	-65	NA	NA
	---	X	X		NA	-65	NA	NA
				X	-50	-65	NA	NA
Adhesion, min., pounds: tube to ply or cover	X	---	---		10	10	10	10
	---	---	---		NA	8	8	8
	---	X			NA	10	8	8
	---	---			NA	8	6	6
				X	10	10	10	10
Ply to ply	X	---	---		10	10	10	10
					NA	8	8	8
					NA	8	6	6
Cover to ply				X	10	10	10	10

* NA - Not applicable."

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3.6.1, after "Month and year of manufacture" add "MS part number (see MS52130)."

3.6.2, after "Month and year of manufacture" add "MS part number (see MS51008)."

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4.3.5.2 thru 4.3.6.1, delete in their entirety and substitute:

"4.3.5.2 Classification of defects. For examination purposes, defects shall be classifed as follows:

Classification of defects.

Categories	Defect	Method of Inspection
Critical	None defined	
Major:	AQL (2.5 percent)	
101	Parts improperly cleansed (see 5.1.1 and 5.2.1.1)	Visual
102	Improper wrap (see 5.1.1 and 5.2.1.1)	Visual
103	Wrap improperly secured (see 5.1.1 and 5.2.1.1)	Visual
104	Improper interior container (see 5.1.1 and 5.2.1.1)	Visual
105	Improper exterior container (see 5.1.1 and 5.2.1.1)	Visual
106	Exterior container improperly sealed (see 5.1.1 and 5.2.1.1)	Visual
107	Some coiled in an unsafe diameter (see 5.1.1 and 5.2.1.1)	Visual
108	Illegible marking (see 5.1.1 and 5.2.2)	Visual
109	Incorrect marking (see 5.1.1 and 5.2.2)	Visual
Minor:	None defined	

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4.3.6 Testing inspection for packages and packs.

4.3.6.1 Quality conformance testing. When level A or B is specified (see 5.1), the samples selected in accordance with 4.3.4.3 shall be tested as specified in 4.5.13, 4.5.14, and 4.5.15."

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4.5.6, delete first five lines on page 22 and substitute "1/3 by volume, alcohol."

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Add new paragraph 4.5.15, as follows:

"4.5.15 Packaging, packing, and marking inspection (for Army use only). Packaging, packing and marking shall be examined as specified in 4.3.5.2 and tested to determine conformance to 5.1.1."

5. thru 5.1.2.2, delete in their entirety and substitute:

"5. PACKAGING

5.1 Packaging, packing and marking (for Army use only).

5.1.1 Packaging, packing and marking. Cleaning, packaging, packing and marking shall be in accordance with applicable packaging data sheet, or other applicable documents specified by the procuring activity. In addition to marking called for on specified documents, interior packages and shipping containers shall include the date of manufacture, or date vulcanized, for the packaged items.

5.2 Packaging, packing, and marking (for civil agencies and Non-Army Military activities).

5.2.1 Packaging and packing levels. Packaging and packing shall be level A, B or commercial as specified (see 6.2).

5.2.1.1 Packaging and packing. Cleaning, packaging and packing shall conform to MIL-R-775. When tested as specified in 4.5.13, the package shall meet requirements specified in MIL-P-116 for the method of packaging used. When tested as specified in 4.5.14, level A packs shall meet the requirements of the interior packing test of MIL-STD-1186. Level B packs shall meet the requirements of the same test, except the height of the drop shall be 75 percent of the height of the drop specified therein.

5.2.2 Marking.

5.2.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with FED-STD-123.

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5.2.2.2 Military activities (Non-Army). In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-H775."

The margin of this amendment is marked with a vertical line to indicate where the change from the previous amendment was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notation and relationship to the last previous amendment.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - YD

Review Activities:
Navy - YD
DLA - CS

User interest:
Army - ME

Preparing activity:
Army - AT

CIVIL AGENCY COORDINATING ACTIVITIES:

NASA - JFK
GSA - FSS
HSS - FEC
DOT - ACO
USDA - APS
TREASURY - ATF

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